

Mohammad Orabe

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Machine learning researcher & engineer specializing in real-time multimodal deep learning, computer vision, open-source development, and end-to-end ML frameworks. Portfolio available on website.

Education

M.Sc. in Computer Science

Feb 2022 – Present

Technical University of Berlin (TU Berlin)

Thesis: Real-Time Brain Activity Recognition Using Deep Learning

Developed deep learning models (LSTM, Transformer) for real-time activity recognition from multimodal sensor data (IMU, EMG, EEG), with emphasis on sequential modeling and robust embedded application.

B.Sc. in Computer Science

Oct 2018 – Sep 2021

TU Berlin

Thesis: Pose Estimation and Multi-Object Tracking with Deep Learning

Designed and implemented a CNN-based vision pipeline with sensor fusion for identity-preserving multi-object tracking in complex, real-world environments.

Experience

Research Assistant – Machine Learning

Feb 2025 – Present

Quality in AI Group, PTB – National Metrology Institute of the Federal Republic of Germany

Developed a framework for uncertainty evaluation in inverse machine learning methods. Built an end-to-end pipeline for data simulation, modeling, and visualization.

Research Assistant – Signal Processing

Jun 2023 – Dec 2024

Brain and Data Science Group, Charité – University Medicine Berlin

Implemented and integrated scalable methods for time series preprocessing and multivariate signal analysis into standard scientific libraries.

Research Assistant – ML & Data Analytics

Jul 2021 – Jun 2023

Winter Lab, Humboldt University of Berlin

Built modular machine learning pipelines and statistical toolboxes for behavioral tracking environments. Applied deep learning for object detection and developed reproducible analytics workflows.

Technical Support – Systems & Automation

Feb 2017 – Jul 2020

Institute for International Communication (IHK), Berlin

Managed Linux/Windows systems, diagnosed and resolved technical issues, and automated administrative tasks using Bash scripting.

Open-Source Softwares – Lead Developer (Selected)

CaliBrain – Calibrated Source Imaging (Python)

Developed a benchmarking framework for uncertainty evaluation and calibration in inverse machine learning methods.

GaitMod – Real-Time Gait Modulation Detection (Python)

Developed a library for real-time gait activity recognition using multimodal sensor data (IMU, LFP, EMG, EEG) and deep learning models (LSTM, CNN, Transformer).

GroupMouseTrack – Multi-Mouse Tracking & Pose Estimation (Python)

Built a CNN-based toolkit for pose estimation and identity-preserving tracking of visually identical entities with RFID-based correction.

ColonyRack – Spatiotemporal Data Analysis Toolbox (R)

Developed a library to process and analyze behavioral data from large-scale systems. Mapped social dynamics using continuous spatiotemporal sensor streams.

Projects (Selected)

MRI Segmentation and Classification – UNet/SegResNet for Brain Imaging

Implemented deep learning models for MRI-based brain tumor segmentation and classification.

Multimodal Variational Autoencoder (MVAE) – Joint Latent Representation

Implemented a generative model for multi-modal learning and image-label translation using an M-VAE.

MNE-Python Contributions – Connectivity Methods Integration

Implemented multivariate connectivity methods to a core module to support signal processing workflows.

Neurolib Integration – Simulation Model Expansion

Integrated standardized atlases and lead-field matrices into neurolib to support scalable brain simulations.

Teaching & Mentorship

Founder & Lead Instructor, MathCodeLab

2023 – Present

Launched a tech education platform offering in-person & online courses in programming and ML. Taught 10+ semester-long courses to over 50 students and built a community of over 240 members.

Teaching Assistant – Biomedical Data Science, TU Berlin

Winter 2024/2025

Teaching a Master's-level seminar on advanced machine learning methods in biomedical data science.

Technical Skills

Programming Languages: Python, R, Julia, MATLAB, Java, C/C++

ML Architectures & Models: CNN, RNN (LSTM); Transformers (LLMs); Generative: GANs, AEs; Reinforcement Learning

ML Libraries & Frameworks: TensorFlow, Keras, PyTorch, scikit-learn, NumPy, pandas, SciPy, Numba, OpenCV, matplotlib, seaborn

Applied ML Domains: NLP, Computer Vision, Time Series, Signal Processing

Cloud & DevOps: Docker, AWS, Azure, CI/CD, RESTful APIs

Data Engineering: SQL, ETL Pipelines, Apache Kafka, Apache Spark

Software Development Tools: Git, GitHub, Bash, SSH, gdb, Unit Testing, API Design, Open-Source

Languages

English (Fluent), German (Professional)

Publications & Talks

Orabe, M., & Haufe, S. (In preparation). Detecting gait modulation from multimodal sensor data using deep learning for real-time activity monitoring.

Ismail Huseynov, I., **Orabe, M.**, Haufe, S., Nagarajan, S., & Hashemi, A. (In Preparation). Investigating the uncertainty estimation of inverse solvers in EEG/MEG brain source imaging.

Binns, T.S., **Orabe, M.**, Nguyen, T.D., Köhler, R.M., Pellegrini, F., & Haufe, S. (2024). Multivariate connectivity methods in the MNE-Python toolbox. *Neural Traces, Berlin, Germany. (Software Poster)*

References

Available upon request.